

Roll No.

TAE-501

B. TECH. (ME) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
ELECTRIC VEHICLE TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

(iii) Draw neat diagrams as necessary.

(iv) Make assumptions for any missing data.

1. (a) Derive an expression for the rolling resistance force of an automobile.

(CO1)

OR

- (b) What is meant by vehicle performance curve ? Explain with a relevant example.

(CO1)

2. (a) Explain all the forces that need to be considered in the tractive effort of an automobile.

(CO1)

OR

- (b) Compare the merits and demerits of series and parallel hybrid vehicles.

(CO1)

P. T. O.

(2)

3. (a) Explain the redox reactions taking place in a Lithium ion cell. (CO2)

OR

- (b) Describe the redox reactions taking place in a Lead Acid cell. (CO2)

4. (a) Compare the advantages and disadvantages of different types of fuel cells. (CO2)

OR

- (b) Explain the construction and operation of a fuel cell. (CO2)

5. (a) Describe the permanent magnet motor drives. (CO3)

OR

- (b) Explain about induction motor drives. (CO3)